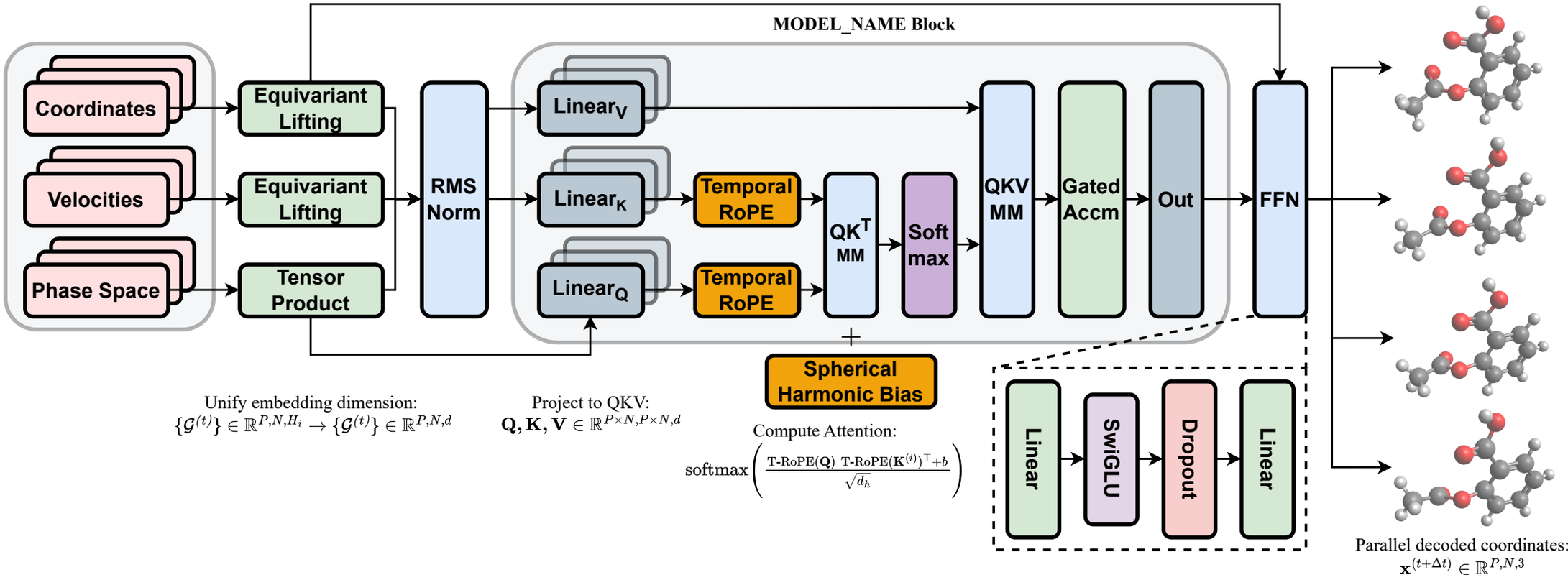
MODEL_NAME Block
Coordinates
Velocities
Phase Space
Equivariant Lifting
Equivariant Lifting
Tensor Product
RMS Norm
Linear$_V$
Linear$_K$
Linear$_Q$
Temporal RoPE
Temporal RoPE
QK^T MM
Soft max
QKV MM
Gated Accm
Out
FFN
Spherical Harmonic Bias
Linear
SwiGLU
Dropout
Linear
Unify embedding dimension:
$\{\mathcal{G}^{(t)}\} \in \mathbb{R}^{P,N,H_i} \rightarrow \{\mathcal{G}^{(t)}\} \in \mathbb{R}^{P,N,d}$
Project to QKV:
$\mathbf{Q},\mathbf{K},\mathbf{V} \in \mathbb{R}^{P\times N,P\times N,d}$
Compute Attention:
$\text{softmax}\left(\frac{\text{T-RoPE}(\mathbf{Q})\ \text{T-RoPE}(\mathbf{K}^{(i)})^\top + b}{\sqrt{d_h}}\right)$
Parallel decoded coordinates:
$\mathbf{x}^{(t+\Delta t)} \in \mathbb{R}^{P,N,3}$